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Note I.2 — Invariance of Plurigenera

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These notes give a self-contained account of Păun’s proof [Pău07] of the deformation invariance of plurigenera for smooth projective families, following the presentation of Demailly [Dem12]. The central idea is to reduce the extension of pluricanonical sections to the construction of a limiting singular Hermitian metric via an inductive application of the Ohsawa–Takegoshi L^2 extension theorem.

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1 Introduction

Let $f: X \rightarrow \Delta$ be a smooth projective family over the unit disk, with smooth fibers $X_t = f^{-1}(t)$. The m -th plurigenus of a fiber is

$$P_m(X_t) = \dim H^0(X_t, \mathcal{O}_{X_t}(mK_{X_t})).$$

The deformation invariance of plurigenera asserts that $t \mapsto P_m(X_t)$ is constant. Upper semi-continuity is standard; the non-trivial direction is showing that sections over the special fiber X_0 extend to the total space.

Siu [Siu98] first proved invariance for families of varieties of general type, using a careful inductive argument with L^2 methods. A key observation in his proof is that extending a section $\sigma \in H^0(X_0, \mathcal{O}(mK_{X_0}))$ can be reformulated as constructing a suitable singular Hermitian metric on the pluricanonical bundle, to which the Ohsawa–Takegoshi theorem can be applied. Subsequent work [Siu02] extended the method to more general twisted settings.

Păun [Pău07] gave a clean *one-tower proof* that works for all smooth projective families. His strategy is to:

1. insert an auxiliary ample line bundle A and build, inductively, a tower of line bundles $F_p \approx p(K_X + \frac{1}{m}L) + A$ carrying global sections with uniform L^2 control;
2. extract a limiting psh weight ψ on $K_X + \frac{1}{m}L$ by a Hölder–Jensen compactness argument;
3. apply Ohsawa–Takegoshi once more, using the limiting metric, to extend σ .

The key slogan is: *convert the section extension problem into a metric construction problem, since metrics can be rescaled and limits can be taken.*

Convention. Throughout, $\Delta \subset \mathbb{C}$ is the unit disk, $X_0 = f^{-1}(0)$ is the central fiber, and all line bundles are holomorphic. We write $h = e^{-\varphi}$ for a singular Hermitian metric h with local psh weight φ . The *multiplier ideal sheaf* of h is

$$\mathcal{I}(h) = \mathcal{I}(\varphi) = \{f \in \mathcal{O}_X \mid |f|^2 e^{-\varphi} \text{ is locally } L^1\}.$$

A singular Hermitian metric h on a line bundle L is *semi-positive* if its curvature current $i\Theta_{L,h} = i\partial\bar{\partial}\varphi \geq 0$ in the sense of currents.

2 The Ohsawa–Takegoshi Extension Theorem

All extension arguments in these notes rest on the following result.

Theorem 2.1 (Ohsawa–Takegoshi–Manivel, [Dem12, Lemma 17.3]). Let $\pi: X \rightarrow \Delta$ be a smooth projective family over the unit disk and let (L, h) be a singular Hermitian line bundle on X with semi-positive curvature current $i\Theta_{L,h} \geq 0$. Let ω be a global Kähler metric on X and let dV_X, dV_{X_0} be the induced volume forms. Assume $h|_{X_0}$ is well defined.

Then any section $u \in H^0(X_0, (K_X + L)|_{X_0} \otimes \mathcal{J}(h|_{X_0}))$ extends to a section $\tilde{u} \in H^0(X, K_X + L)$ with L^2 estimate

$$\int_X \|\tilde{u}\|_{\omega \otimes h}^2 dV_X \leq C_0 \int_{X_0} \|u\|_{\omega \otimes h}^2 dV_{X_0},$$

where $C_0 \geq 0$ is a *universal* constant independent of X , L , and h .

Remark 2.2. The universality of C_0 is crucial. In the inductive procedure of Section 3, we apply Theorem 2.1 at each step with a different metric, but the same bound C_0 holds throughout. This is what ultimately gives a uniform bound on the induced psh weights ψ_p .

Remark 2.3. The condition $u \in H^0(X_0, (K_X + L)|_{X_0} \otimes \mathcal{J}(h|_{X_0}))$ is equivalent to the L^2 integrability $\int_{X_0} \|u\|_h^2 dV_{X_0} < +\infty$.

3 Păun's Extension Theorem

3.1 Statement

Theorem 3.1 (Păun's extension theorem, [Dem12, Theorem 17.2]). Let $\pi: X \rightarrow \Delta$ be a smooth projective family over the unit disk. Let (L, h) be a singular Hermitian line bundle on X with semi-positive curvature current $i\Theta_{L,h} \geq 0$, and assume $h|_{X_0}$ is well defined. Then the restriction map

$$H^0(X, m\mathcal{O}_X(K_X + L)) \longrightarrow H^0(X_0, \mathcal{O}_{X_0}(m(K_X + L)) \otimes \mathcal{J}(h|_{X_0}))$$

is surjective. In other words, every section $\sigma \in H^0(X_0, \mathcal{O}(m(K_X + L)) \otimes \mathcal{J}(h|_{X_0}))$ extends to X .

3.2 Proof strategy

Proof Idea 3.2. Write $h = e^{-\varphi_L}$. The proof divides into four steps.

Step 1 (Tower of sections). Fix an auxiliary ample A . Define line bundles

$$F_0 = A, \quad F_k = F_{k-1} + K_X \quad (1 \leq k \leq m-1), \quad F_k = F_{k-1} + K_X + L \quad (k \equiv 0 \pmod{m}).$$

Use Ohsawa–Takegoshi inductively to extend sections from X_0 to each F_p with uniform L^2 control.

Step 2 (Limiting metric). Show the psh weights $\psi_p = \frac{1}{p} \log \sum_i |\tilde{u}_i^{(p)}|^2$ are locally uniformly bounded above via Hölder and the mean value inequality. Extract a limiting psh weight ψ on $K_X + \frac{1}{m}L$.

Step 3 (Properties of limit). Verify $\psi|_{X_0} \geq \frac{1}{m} \log |\sigma|^2 \not\equiv -\infty$ and that the limiting metric $H = e^{-m\psi}$ satisfies $\|\sigma\|_H \leq 1$.

Step 4 (Extension). Equip $G = (m-1)K_X + L$ with the metric $\gamma = H^{1-1/m} \cdot h^{1/m}$, verify the L^2 integrability of σ with respect to γ , and apply Ohsawa–Takegoshi.

3.3 Step 1: The tower of auxiliary line bundles and sections

Fix an auxiliary ample line bundle A on X , to be taken sufficiently ample. Define a sequence of line bundles inductively by $F_0 = A$ and

$$F_p = \begin{cases} F_{p-1} + K_X + L & \text{if } p \equiv 0 \pmod{m}, \\ F_{p-1} + K_X & \text{if } p \not\equiv 0 \pmod{m}. \end{cases} \quad (1)$$

Explicitly, $F_p = A + pK_X + \lfloor p/m \rfloor L$, so as $\mathbb{Q}$ -line bundles,

$$\frac{1}{p} F_p = K_X + \frac{\lfloor p/m \rfloor}{p} L + \frac{1}{p} A \xrightarrow{p \rightarrow \infty} K_X + \frac{1}{m} L.$$

In particular $F_0, F_1, \dots, F_{m-1} = A, K_X + A, \dots, (m-1)K_X + A$ contain no L -twist, and $F_m = mK_X + L + A$.

We inductively construct families of global sections $\{\tilde{u}_i^{(p)}\}_{i=1}^N$ of F_p over X , satisfying:

- (a) For $0 \leq p \leq m-1$, the sections $\{\tilde{u}_i^{(p)}\}$ generate F_p globally (achieved by taking A sufficiently ample).
- (b) Writing $p = mq + r$ with $0 \leq r \leq m-1$, the restriction to X_0 is $u_i^{(p)} := \tilde{u}_i^{(p)}|_{X_0} = \sigma^q u_i^{(r)}$, where $u_i^{(r)} = \tilde{u}_i^{(r)}|_{X_0}$.

Condition (b) is the key m -periodicity: at level $p = mq + r$ the section on X_0 is simply the base section $u_i^{(r)}$ tensored with q copies of σ .

The base cases $p = 0, \dots, m-1$ are given by property (a). For the induction step at $p \geq m$, assume $\{\tilde{u}_i^{(p-1)}\}$ of F_{p-1} are constructed. We equip F_{p-1} with the *tautological metric*

$$h_{p-1} = \frac{|\xi|^2}{\sum_i |\tilde{u}_i^{(p-1)}(x)|^2}.$$

Since $\{\tilde{u}_i^{(p-1)}\}$ generate F_{p-1} when $p-1 \leq m-1$ (and more generally the sections are globally defined), the tautological metric is smooth with semi-positive curvature. We then apply Theorem 2.1 to extend $u_i^{(p)}$ from X_0 to X , with the bundle

$$L' = \begin{cases} F_{p-1} + L & \text{if } p \equiv 0 \pmod{m}, \\ F_{p-1} & \text{if } p \not\equiv 0 \pmod{m}, \end{cases}$$

so that $K_X + L' = F_p$ in both cases, and the metric $h_{L'} = h_{p-1} \cdot h$ or $h_{L'} = h_{p-1}$ accordingly.

3.4 Step 2: Verifying the L^2 condition

To apply Theorem 2.1 at each step we must verify the L^2 condition on X_0 . The pointwise norm of $u_i^{(p)}$ with respect to $h_{L'}$ is proportional, in local trivializations, to the ratio

$$\frac{|u_i^{(p)}|^2}{\sum_j |u_j^{(p-1)}|^2} \cdot \begin{cases} e^{-\varphi_L} & p \equiv 0 \pmod{m}, \\ 1 & \text{otherwise.} \end{cases}$$

The m -periodicity condition (b) gives $u_i^{(p)} = \sigma^q u_i^{(r)}$ for $p = mq + r$, so the sum over i reduces to:

$$\frac{\sum_i |u_i^{(p)}|^2}{\sum_j |u_j^{(p-1)}|^2} \cdot e^{-\varphi_{L'}} = \begin{cases} \frac{\sum_i |u_i^{(r)}|^2}{\sum_j |u_j^{(r-1)}|^2} & \text{if } p = mq + r, 0 < r \leq m-1, \\ \frac{\sum_i |u_i^{(0)}|^2}{\sum_j |u_j^{(m-1)}|^2} |\sigma|^2 e^{-\varphi_L} & \text{if } p \equiv 0 \pmod{m}. \end{cases} \quad (2)$$

In the first case, $\{u_i^{(r)}\}$ generate $F_r|_{X_0}$ so the ratio is smooth and bounded on X_0 , hence integrable. In the second case, $|\sigma|^2 e^{-\varphi_L}$ is locally integrable on X_0 since $\sigma \in \mathcal{J}(h|_{X_0})$ by hypothesis.

Therefore there exists a constant $C_1 \geq 0$, *independent of p* , such that

$$\int_{X_0} \frac{\sum_i |u_i^{(p)}|^2}{\sum_j |u_j^{(p-1)}|^2} \cdot e^{-\varphi_{L'}} dV_\omega \leq C_1 \quad \text{for all } p \geq 1. \quad (3)$$

By Theorem 2.1, each $u_i^{(p)}$ extends to $\tilde{u}_i^{(p)} \in H^0(X, F_p)$ with L^2 estimate

$$\int_X \frac{\sum_i |\tilde{u}_i^{(p)}|^2}{\sum_j |\tilde{u}_j^{(p-1)}|^2} \cdot e^{-\varphi_{L'}} dV_\omega \leq C_2 := C_0 C_1. \quad (4)$$

Remark 3.3. The uniformity of C_2 across all p is the key input. It comes from the universality of C_0 in Theorem 2.1 and from the fact that (2) involves only finitely many pieces of data (the base sections $u_i^{(0)}, \dots, u_i^{(m-1)}$ and σ), so C_1 is independent of p .

3.5 Step 3: Construction of the limiting metric

Define the *tautological psh weight*

$$\psi_p := \frac{1}{p} \log \sum_i |\tilde{u}_i^{(p)}|^2,$$

a semi-positive metric weight on the $\mathbb{Q}$ -line bundle $\frac{1}{p}F_p$. We want to show that $\{\psi_p\}$ is locally uniformly bounded from above so that a convergent subsequence can be extracted.

Set

$$A_p := \frac{\sum_i |\tilde{u}_i^{(p)}|^2}{\sum_i |\tilde{u}_i^{(p-1)}|^2} \quad (p \geq 2), \quad A_1 := \sum_i |\tilde{u}_i^{(1)}|^2.$$

By telescoping, $\sum_i |\tilde{u}_i^{(p)}|^2 = A_p \cdot A_{p-1} \cdots A_1$, so

$$\left(\sum_i |\tilde{u}_i^{(p)}|^2 \right)^{1/p} = A_p^{1/p} \cdot A_{p-1}^{1/p} \cdots A_1^{1/p}.$$

Applying Hölder's inequality with all exponents equal to p :

$$\int_X \left(\sum_i |\tilde{u}_i^{(p)}|^2 \right)^{1/p} dV_\omega = \int_X A_p^{1/p} \cdots A_1^{1/p} dV_\omega \leq \prod_{k=1}^p \left(\int_X A_k dV_\omega \right)^{1/p}.$$

By the L^2 estimate (4) from Ohsawa–Takegoshi, $\int_X A_p dV_\omega \leq C_2$ for all $p \geq 2$. The base term satisfies $\int_X A_1 dV_\omega \leq C_3$. Therefore

$$\int_X \left(\sum_i |\tilde{u}_i^{(p)}|^2 \right)^{1/p} dV_\omega \leq C_2^{1-1/p} \cdot C_3^{1/p} \leq C_4, \quad (5)$$

a constant independent of p .

Since ψ_p is psh, the mean value inequality and Jensen's inequality give, for any ball $B(z_0, r)$:

$$\psi_p(z_0) \leq \frac{1}{|B(z_0, r)|} \int_{B(z_0, r)} \log \left(\sum_i |\tilde{u}_i^{(p)}|^2 \right)^{1/p} dV \leq \frac{1}{|B(z_0, r)|} \log \int_{B(z_0, r)} \left(\sum_i |\tilde{u}_i^{(p)}|^2 \right)^{1/p} dV \leq \frac{\log C_4}{|B(z_0, r)|},$$

which is locally uniformly bounded from above, independently of p .

By the compactness theorem for locally uniformly bounded-above psh functions, there exists a subsequence $p_\nu \rightarrow \infty$ such that ψ_{p_ν} converges in L^1_{loc} to the psh function

$$\psi_\infty(z) = \limsup_{\zeta \rightarrow z} \lim_{\nu \rightarrow \infty} \psi_{p_\nu}(\zeta),$$

which is a semi-positive metric weight on $K_X + \frac{1}{m}L$.

3.6 Step 4: Properties of the limiting metric

Since $\tilde{u}_i^{(p)}|_{X_0} = \sigma^q u_i^{(r)}$ for $p = mq + r$, we have on X_0 :

$$\frac{1}{p} \log \sum_i |u_i^{(p)}|^2 = \frac{1}{p} \log \left(|\sigma|^{2q} \sum_i |u_i^{(r)}|^2 \right) \xrightarrow{p \rightarrow \infty} \frac{1}{m} \log |\sigma|^2 \not\equiv -\infty.$$

Therefore $\psi|_{X_0} \geq \frac{1}{m} \log |\sigma|^2 \not\equiv -\infty$, so the metric $H := e^{-m\psi}$ on $m(K_X + \frac{1}{m}L) = mK_X + L$ is well defined on X_0 . Moreover,

$$\|\sigma\|_H^2 = |\sigma|^2 e^{-m\psi} \leq |\sigma|^2 \cdot e^{-\log |\sigma|^2} = 1,$$

so $\|\sigma\|_H \leq 1$.

3.7 Step 5: Extending σ via Ohsawa–Takegoshi

Write $m(K_X + L) = K_X + G$ with $G = (m-1)K_X + L$. Equip G with the metric

$$\gamma = H^{1-1/m} \cdot h^{1/m}.$$

Since H and h are semi-positive, γ is semi-positive on X . It remains to check the L^2 condition for σ on X_0 .

In local trivializations, using $\|\sigma\|_H \leq 1$ in the second step:

$$\|\sigma\|_{\omega \otimes \gamma}^2 \leq C |\sigma|^2 \exp(-(m-1)\psi - \frac{1}{m}\varphi_L) = C \|\sigma\|_H^{2(1-1/m)} \cdot |\sigma|^{2/m} e^{-\varphi_L/m} \leq C (|\sigma|^2 e^{-\varphi_L})^{1/m}.$$

Since $|\sigma|^2 e^{-\varphi_L}$ is locally L^1 on X_0 (as $\sigma \in \mathcal{J}(h|_{X_0})$), the right-hand side is locally L^1 as well. Thus

$$\sigma \in H^0\left(X_0, (K_X + G)|_{X_0} \otimes \mathcal{J}(\gamma|_{X_0})\right),$$

and a final application of Theorem 2.1 to (G, γ) extends σ to X . $\square$

4 Invariance of Plurigenera

Corollary 4.1 (Invariance of plurigenera, Păun [Pău07]). Let $f: X \rightarrow \Delta$ be a smooth projective family over the unit disk. Then $P_m(X_t) = \dim H^0(X_t, \mathcal{O}(mK_{X_t}))$ is independent of t .

Proof. Upper semi-continuity of P_m is classical. For lower semi-continuity, it suffices to show every section $\sigma \in H^0(X_0, \mathcal{O}(mK_{X_0}))$ extends to X .

Apply Theorem 3.1 with $L = \mathcal{O}_X$ and $h = 1$ (the trivial metric). The curvature is zero (hence semi-positive), $h|_{X_0}$ is well defined, and $\mathcal{J}(h|_{X_0}) = \mathcal{O}_{X_0}$ is trivial. The section σ satisfies the L^2 condition automatically since X_0 is compact. Theorem 3.1 then provides the extension. $\square$

Remark 4.2. Siu [Siu98] first proved Corollary 4.1 for families of general type, and later [Siu02] extended it to twisted pluricanonical systems. Păun’s proof removes the general type hypothesis with an argument that is also considerably shorter.

5 Further Remarks

5.1 Siu’s conjecture and the non-projective setting

Theorem 3.1 is stated for projective families. Projectivity is used to guarantee global generation of $F_0, \dots, F_{m-1}$ by a sufficiently ample A , and to ensure the compactness argument for psh functions applies in the correct cohomology class.

For Kähler families (non-projective), the ample bundle A does not exist, and a substitute must be found. This is related to Siu’s conjecture that plurigenera are invariant for all smooth families of compact complex manifolds. The question is open in general; partial results are available via the MMP in the Kähler setting [DHP24].

Remark 5.1. Nakayama [Nak04] proved lower semi-continuity of plurigenera for families admitting a relative minimal model, providing an algebraic complement to the analytic approach.

5.2 The role of the auxiliary ample bundle

The insertion of A is the mechanism by which Păun “squeezes extra positivity” into the problem. The $\mathbb{Q}$ -line bundles $\frac{1}{p}F_p = K_X + \frac{\lfloor p/m \rfloor}{p}L + \frac{1}{p}A$ converge to $K_X + \frac{1}{m}L$ from the positive side in $H^{1,1}(X, \mathbb{R})$. This positivity ensures that $F_0, \dots, F_{m-1}$ are globally generated (so the tautological metrics are smooth) and that the limiting psh weight lies in a bounded family within the Kähler cone.

5.3 Relation to the minimal model program

Kawamata [Kaw99] gave a proof of invariance of plurigenera using the subadjunction formula for log canonical centers. Takayama [Tak07] extended the result to broader settings using multiplier ideal sheaves.

In the Kähler setting, the recent work of Das–Hacon–Păun [DHP24] establishes key parts of the MMP for compact Kähler manifolds. The analytic tools from Păun’s one-tower proof — tautological metrics, limiting arguments, and uniform L^2 estimates — play a central role in that program.

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